Seongyong **Park**

Postdoctoral Research Fellow · Cancer Data Science Lab (CDSL)

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"Things which we see are not by themselves what we see... It remains completely unknown to us what the objects may be by themselves and apart from the receptivity of our senses. We know nothing but our manner of perceiving them." ***Immanuel Kant***

Summary

I am a computational scientist and AI researcher developing machine learning and statistical methods for biomedical discovery. I am currently a Postdoctoral Research Fellow at the Cancer Data Science Laboratory (CDSL), NCI/CCR, NIH, where my research focuses on computational immuno-oncology and the inference of cell–cell communication. In particular, I develop methods to infer the activity of secreted proteins and characterize how signaling is coordinated across cells and tissues using bulk, single-cell, and spatial transcriptomics. This work integrates biological knowledge, statistical modeling, and machine learning to uncover signaling mechanisms, identify clinically relevant biomarkers, and improve our understanding of therapeutic response.

My research has led to computational frameworks and software for biological activity inference, including SecAct, a framework for inferring secreted-protein signaling activities across more than 1,000 human secreted proteins, and its Python implementation, as well as GPU-accelerated tools for large-scale spatial analysis. I have also developed machine learning approaches for clinical biomarker discovery, including IC2Bert for immune checkpoint blockade response prediction, and contributed to broader studies of artificial intelligence in precision oncology. My work has been published in *Nature Methods*, *Nature Cancer*, *Scientific Reports*, and other peer-reviewed journals, and has resulted in open-source software and three issued patents.

More broadly, my methodological interests include biologically informed machine learning, representation learning, foundation and self-supervised learning, sparse modeling, and statistical methods for evaluating biological data and model quality. Before focusing on computational immuno-oncology, I worked on gene-expression-based disease and drug-response prediction, biosignal processing and diagnostic modeling, and microfluidic systems for synthetic biology. This interdisciplinary training, spanning mechanical engineering, bioengineering, computational biology, and AI, allows me to approach biomedical problems from both methodological and biological perspectives and to develop computational approaches that translate across data modalities and research domains.

Work Experience

Cancer Data Science Laboratory (CDSL), NCI/CCR, NIH — *Bethesda, MD, USA*

Postdoctoral Research Fellow — *Jan. 2023 – Present*

- Develop computational methods for inferring cell–cell communication and secreted-protein/cytokine signaling from bulk, single-cell, and spatial transcriptomics, with applications in cancer and immuno-oncology.
- Developed **SecAct**, a framework for inferring the activities of 1,170 human secreted proteins and characterizing their spatial and cellular signaling patterns; published in *Nature Methods* (2026).
- Develop and maintain open-source software for large-scale signaling inference, including **secactpy** (Python), **spatial-gpu** (GPU-accelerated spatial analysis), and contributions to the Jiang Lab R packages **secact** and **spacet**.
- Develop biologically informed machine-learning approaches for clinical biomarker discovery and treatment-response prediction, including **IC2Bert**, a gene-expression foundation model for immune checkpoint blockade response prediction, published in *Scientific Reports* (2025).
- Develop computational approaches for analyzing high-dimensional and spatially resolved tumor data to identify signaling programs associated with tumor–immune interactions and therapeutic response.
- Co-authored a *Nature Cancer* (2025) review examining the role and emerging applications of artificial intelligence in precision oncology.

BAPOR Lab, AMC — *Seoul, South Korea*

Postdoctoral Researcher — *Aug. 2022 – Dec. 2022*

- Developed deep-learning and signal-processing methods for automated diagnosis and prognostic modeling using physiological biosignals, with a focus on phonocardiograms and other acoustic signals.
- Designed temporal convolutional and multimodal feature-learning approaches for extracting clinically relevant information from noisy and heterogeneous biosignal recordings.
- Developed machine-learning methods for physiological state and disease prediction, contributing to peer-reviewed publications and translational research projects.
- Contributed to technology development resulting in patent applications for automated heart-valve abnormality detection, lung-sound-based pulmonary function assessment, and acoustic estimation of physiological parameters.

Synergistic Bioinformatics Lab, KAIST
Daejeon, South Korea

Ph.D. Research Scholar *Aug. 2015 – Aug. 2022*

- Developed machine-learning and deep-learning methods for biomedical prediction using gene-expression, molecular, and spectroscopic data.
- Developed computational models for disease diagnosis, treatment-response prediction, drug repositioning, and biomolecular classification, integrating biological knowledge with statistical and machine-learning approaches.
- Developed sparse representation and latent-space learning methods for classification of biomolecules and biological sequences, including anticancer peptides and protein-related prediction tasks.
- Developed deep-learning methods for surface-enhanced Raman spectroscopy (SERS) to enable automated biomolecule detection and reduce inter-laboratory variation in spectroscopic measurements.
- Developed statistical and effect-size methods for robust biological data and assay-quality evaluation, including **GSSMD** and **GMDM**.
- Published interdisciplinary research in bioinformatics, computational biology, spectroscopy, and biomedical AI, including first-author work in *Analyst*, *Digital Signal Processing*, *Sensors*, *Biosensors*, *Cancers*, and related journals.

Synergistic Bioinformatics Lab, KAIST
Daejeon, South Korea

Researcher *Aug. 2014 – Aug. 2015*

- Developed computational methods using graph theory and machine learning for gene-expression-based disease and drug-response prediction.
- Applied network-based and statistical modeling approaches to identify molecular features and gene groups associated with disease phenotypes and therapeutic response.
- Contributed to research projects leading to peer-reviewed publications and intellectual property in computational biomedical analysis.

Biobrain Inc.
Daejeon, South Korea

Researcher *Aug. 2012 – Aug. 2014*

- Developed signal-processing and machine-learning algorithms for analysis of EEG and other physiological biosignals.
- Designed computational pipelines for extracting informative features from noisy biomedical signals for diagnostic and physiological-state modeling.
- Contributed to algorithm development, experimental evaluation, technical reporting, and translation of computational methods into biomedical applications.

Planning and Evaluation Office, Korea Institute of Energy Technology Evaluation and Planning (KETEP)
Seoul, South Korea

Researcher *Feb. 2012 – Aug. 2012*

- Evaluated government-funded research projects in renewable energy and reviewed technical progress and research outcomes.
- Prepared research trend reports and technical documentation to support project evaluation and planning.
- Supported administrative coordination and reporting for research programs.

Skills

Research	Cell-to-cell communication, cytokine / secreted-protein activity inference, single-cell & spatial transcriptomics, biological-prior-informed ML, sparse representation learning, overlap statistics, biosignal processing.
Programming	Python, R, MATLAB, LaTeX, Shell/SLURM; PyTorch, TensorFlow, Keras, scikit-learn, Bioconductor, Seurat, Scanpy, CuPy/JAX for GPU.
Open-source software	Author: **secactpy** (Python, Jiang Lab), **spatial-gpu** (GPU spatial kernels). Contributor: **secact**, **spacet** (Jiang Lab R stack).
Infrastructure	HPC (SLURM on NIH Biowulf), GPU/CPU cluster management, containerization (Singularity/Apptainer), reproducible pipelines.
Languages	Korean (native), English (professional).

Publications

Peer-Reviewed Journal Articles

1. Beibei Ru, Lanqi Gong, Emily Yang, **Seongyong Park**, George Zaki, Kenneth Aldape, Lalage Wakefield, Peng Jiang. "Inference of secreted protein signaling activities in intercellular communication." ***Nature Methods*** 23(8), 1553–1563 (2026). [website]
2. Jaeman Shin, **Seongyong Park**, Kyucheol Shin, Wooyeong Seo, Hyunseok Kim, Doo-Kwang Kim, Bongjin Moon, Sang Gu Cha, et al. "Temporal convolutional neural network-based feature extraction and asynchronous channel information fusion method for heart abnormality detection in phonocardiograms." ***Computer Methods and Programs in Biomedicine*** 269, 108871 (2025). [code]
3. **Seongyong Park**, Sunho Kim, Peng Jiang. "IC2Bert: masked gene expression pretraining and supervised fine tuning for robust immune checkpoint blockade (ICB) response prediction." ***Scientific Reports*** 15, 28044 (2025). [code]
4. Tian-Gen Chang, **Seongyong Park**, Alejandro A. Schäffer, Peng Jiang, Eytan Ruppin. "Hallmarks of artificial intelligence contributions to precision oncology." ***Nature Cancer*** 6(3), 417–431 (2025).
5. Sang-Wook Lee, **Seongyong Park**, Ji-Yeon Kim, Bongjin Moon, Dongwoo Lee, Jaewon Jang, Wooyeong Seo, Hyunseok Kim, Sung-Hoon Kim, et al. "Impact of preanesthetic blood pressure deviations on 30-day postoperative mortality in non-cardiac surgery patients." ***Journal of Korean Medical Science*** 39(35) (2024).

6. **Seongyong Park**, Abdul Wahab, Muhammad Usman, Imran Naseem, Shujaat Khan. "Artificial intelligence in bioimaging and signal processing." ***Frontiers in Physiology*** 14, 1267632 (2023).
7. **Seongyong Park**, Mohammad Sohail Ibrahim, Abdul Wahab, Shujaat Khan. "GMDM: A generalized multi-dimensional distribution overlap metric for data and model quality evaluation." ***Digital Signal Processing*** 134, 103930 (2023).
8. Ehtisham Fazal, Mohammad Sohail Ibrahim, **Seongyong Park**, Imran Naseem, Abdul Wahab. "Anticancer peptides classification using kernel sparse representation classifier." ***IEEE Access*** 11, 17626–17637 (2023).
9. **Seongyong Park**, Abdul Wahab, Minseok Kim, Shujaat Khan. "Self-supervised learning for inter-laboratory variation minimization in surface-enhanced Raman scattering spectroscopy." ***Analyst*** 148(7), 1473–1482 (2023).
10. **Seongyong Park**, Gwansu Yi. "Development of gene expression-based random forest model for predicting neoadjuvant chemotherapy response in triple-negative breast cancer." ***Cancers*** 14(4), 881 (2022).
11. Muhammad Usman, Shujaat Khan, **Seongyong Park**, Abdul Wahab. "AFP-SRC: identification of antifreeze proteins using sparse representation classifier." ***Neural Computing and Applications*** 34(3), 2275–2285 (2022).
12. **Seongyong Park**, Jaeseok Lee, Shujaat Khan, Abdul Wahab, Minseok Kim. "Machine learning-based heavy metal ion detection using surface-enhanced Raman spectroscopy." ***Sensors*** 22(2), 596 (2022).
13. **Seongyong Park**, Jaeseok Lee, Shujaat Khan, Abdul Wahab, Minseok Kim. "SERSNet: Surface-enhanced Raman spectroscopy based biomolecule detection using deep neural network." ***Biosensors*** 11(12), 490 (2021).
14. Muhammad Usman, Shujaat Khan, **Seongyong Park**, Jeong-A Lee. "AoP-LSE: antioxidant proteins classification using deep latent space encoding of sequence features." ***Current Issues in Molecular Biology*** 43(3), 1489–1501 (2021).
15. Yoon Hyeok Lee, Hojae Choi, **Seongyong Park**, Boah Lee, Gwansu Yi. "Drug repositioning for enzyme modulator based on human metabolite-likeness." ***BMC Bioinformatics*** 18(Suppl 7), 226 (2017).
16. **Seongyong Park**, Xiaoqiang Hong, Woon Sun Choi, Taesung Kim. "Microfabricated ratchet structure integrated concentrator arrays for synthetic bacterial cell-to-cell communication assays." ***Lab on a Chip*** 12(20), 3914–3922 (2012).
17. Vinuselvi Parisutham, **Seongyong Park**, Minseok Kim, Jung Min Park, Taesung Kim, Sung Kuk Lee. "Microfluidic technologies for synthetic biology." ***International Journal of Molecular Sciences*** 12(6), 3576–3593 (2011).
18. Woon Sun Choi, Dogyeong Ha, **Seongyong Park**, Taesung Kim. "Synthetic multicellular cell-to-cell communication in inkjet printed bacterial cell systems." ***Biomaterials*** 32(10), 2500–2507 (2011).
19. **Seongyong Park**, Dasol Kim, Robert J. Mitchell, Taesung Kim. "A microfluidic concentrator array for quantitative predation assays of predatory microbes." ***Lab on a Chip*** 11(17), 2916–2923 (2011).

Conference Proceedings

1. **Seongyong Park**, Shujaat Khan, Muhammad Moinuddin, Ubaid M. Al-Saggaf. "GSSMD: A new standardized effect size measure to improve robustness and interpretability in biological applications." ***IEEE International Conference on Bioinformatics and Biomedicine (BIBM)***, pp. 1096–1099 (2020).

Patents

1. **Seongyong Park**, et al. "Analyzing heart valve abnormalities based on heart murmur." ***KR Patent App. 10-2023-0014972***, registered 30 December 2025.
2. **Seongyong Park**, et al. "Pulmonary function test using lung sound." ***KR Patent App. 10-2023-0022834***, registered 21 February 2023.
3. **Seongyong Park**, et al. "Measuring preload in general anesthesia from an acoustic variability index." ***KR Patent App. 10-2023-0021404***, registered 17 February 2023.
4. Bumki Min, Gwansu Yi, **Seongyong Park**. "Disease prediction from group markers of functionally similar genes." ***KR Patent 10-2236194***, issued 30 March 2021.
5. Taesung Kim, **Seongyong Park**. "Microfluidic concentrator for communication assays of microbes." ***KR Patent 10-1330473***, issued 15 November 2013.
6. Taesung Kim, **Seongyong Park**. "Microfluidic concentrator array for observing predation behavior of microbes." ***KR Patent 10-1238556***, issued 22 February 2013.

Software

Author:

1. **secactpy** — Secreted-protein activity inference from bulk and single-cell expression (Python). github.com/data2intelligence/secactpy
2. **spatial-gpu** — GPU-accelerated spatial kernels for neighborhood activity inference (Python). github.com/psychemistz/spatial-gpu

Contributor (Jiang Lab, NCI):

1. **secact** — Secreted-protein activity inference (R). github.com/data2intelligence/secact
2. **spacet** — Spatial Cellular Estimator for Tumors (R). github.com/data2intelligence/spacet

Education

KAIST(Korea Advanced Institute of Science and Technology) *Daejeon, S.Korea*

Ph.D. in Bio & Brain Engineering *Sep. 2015 - Aug. 2022*

UNIST(Ulsan National Institute of Science and Technology) *Ulsan, S.Korea*

M.S. in Mechanical Engineering *Mar. 2010 - Feb. 2012*

- *Thesis*: Microfluidic Concentrator Array for Quantitative Predation Study of Predatory Microbes[online]

PKNU (Pukyong National University) *Busan, S.Korea*

B.E. in Mechanical Engineering *Mar. 2002 - Feb. 2010*